

Cloud footprint and the road to production

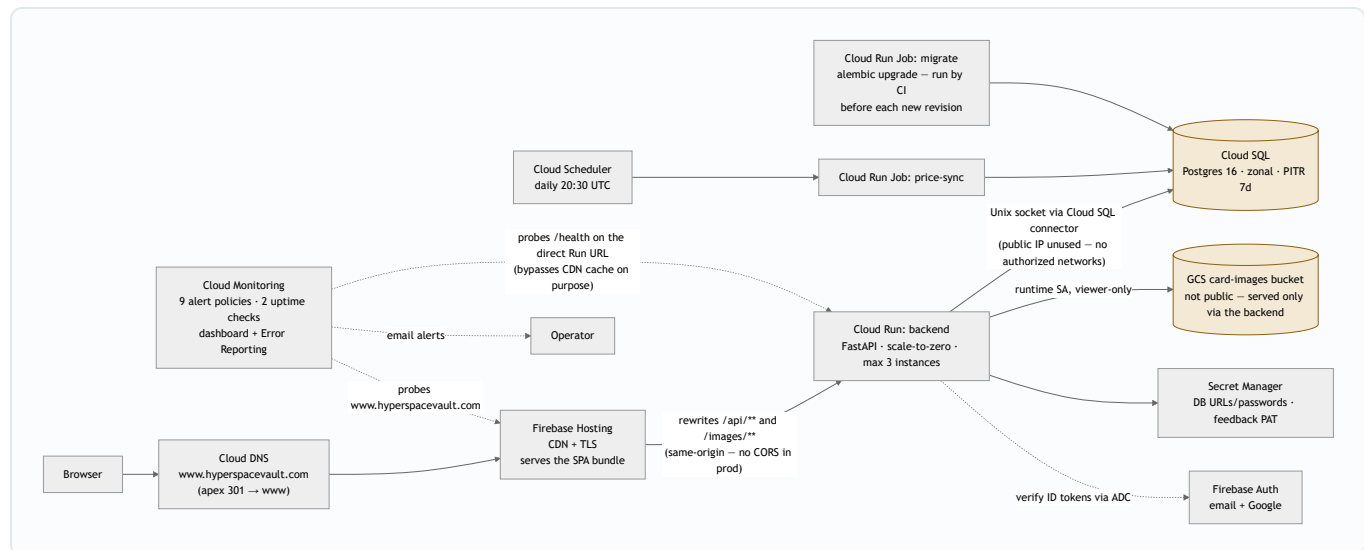
HyperspaceVault · Architecture · View 3 of 3 — Physical

Everything runs on Google Cloud in `us-central1`, in two mirrored environments defined by one shared Terraform module — a dev project and a prod project — plus a frozen sandbox for infra experiments. Firebase Hosting is the CDN and front door; there is deliberately no load balancer.

Prepared 2026-07-28 · Updated 2026-08-16 (v1.4 era: release-gated promotes, applied DB tier, completed restore drills) · Companion views: [Conceptual](#) (domain & actors) · [Logical](#) (components & data)

Production runtime topology

Prod (swu-prod) — request path and managed services



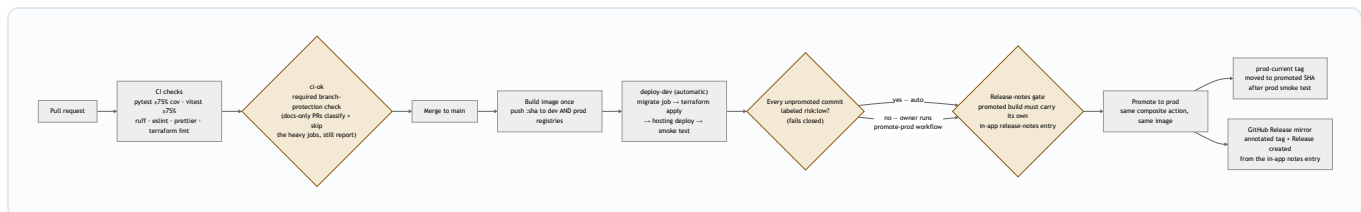
The dev project (`swu-dev-jbapps`) is the same module with smaller knobs — one max instance, smaller database tier, no custom domain. A third project (`swu-sandbox`) exists only so infrastructure experiments never touch, or bill like, production.

Edge caching	Policy	Why
Public catalog reads	<code>max-age=300, s-maxage=3600</code>	Guests hit the CDN, not the backend; $\leq 1h$ staleness accepted
Card images	<code>max-age=31536000, immutable</code>	Object paths are immutable by convention; cache forever
Tenant-scoped routes	<code>private, no-store</code>	Vault data must never land in a shared cache

From pull request to production

Delivery is **build-once, promote-many**: one Docker image is built per merge and pushed to both environments' registries; promotion re-deploys the exact bytes that dev validated. CI authenticates to Google Cloud with workload identity federation — no stored service-account keys anywhere. Database migrations run as a discrete job, to completion, *before* the new service revision takes traffic.

CI/CD — GitHub Actions



- **Human gate by default:** the pipeline ends at dev. Production is a deliberate `workflow_dispatch` with a SHA that must be an ancestor of main.
- **Risk-tiered fast lane:** auto-promotion fires only if *every* commit since the last promote carried a `risk:low` label — a whole-build guard that fails closed on any missing label, missing PR, or unreadable state.
- **prod-current is the source of truth** for "what is production running" — a tag moved only after the prod smoke test passes, readable without any cloud credentials.
- **Every promote is release-noted:** the promote workflow refuses to ship a build whose version lacks its in-app release-notes entry, then mirrors that entry to a GitHub Release on the public repo — users and the repo see the same story.
- **Docs-only changes trigger zero workflow runs** — the pipeline's cost floor is guarded at the trigger level, with change detection as belt-and-braces.

Environments at a glance

	dev — swu-dev-jbapps	prod — swu-prod
Deploys	Automatic on merge to main	Owner-gated promote (or all-risk:low auto)
Front door	swu-dev-jbapps.web.app	www.hyperspacevault.com (apex 301; legacy domain a permanent 301 since 2026-07-30)
Cloud Run max instances	1	3 — bounds worst-case cost of flooding the anonymous catalog endpoints
Cloud SQL tier	db-f1-micro	db-g1-small (applied and live-verified 2026-08-01)
Deletion protection	Off	On
Alerting	8 policies (doubles as a canary for alert config)	9 policies — adds custom-domain uptime

Local development mirrors the cloud shape with Docker Compose: Postgres 16, the backend (auto-migrating on start), the Vite dev server, and a Firebase Auth *emulator* — so a laptop needs **zero GCP credentials**, and row-level security is exercised identically in local, CI, and prod because the same migrations create the same restricted role everywhere.

Cost and operations posture — chosen, not defaulted

- **Scale-to-zero everywhere**, bounded above (3 prod / 1 dev instances), with a \$50/month billing budget alerting at 50/90/100%. The budget is deliberately managed outside Terraform so CI's cloud identity never holds billing permissions.
- **Uptime checks probe the direct backend URL, not the CDN** — a cached front door would happily serve hits long after the backend died. A second prod-only probe covers the real domain so DNS/TLS/Hosting failures aren't invisible either.
- **Zonal database, no HA** — accepted at current scale, with backups + 7-day point-in-time recovery. Stated $RPO \leq 1h$ / $RTO \leq 4h$ — validated: restore drills were performed in both environments (dev rehearsal 2026-07-30, prod drill 2026-08-01) with roughly 26× RTO headroom.
- **Rate limiting is in-memory per instance** (no WAF) — proportionate to current traffic; the instance cap is the backstop against abuse of the anonymous surface.

Known drift, stated on purpose (an as-built diagram should say so): Google sign-in is enabled in the console but not yet declared in Terraform — a tracked backlog item, not a surprise. (Two earlier drift items resolved since this view was first prepared: the prod database tier upgrade was applied 2026-08-01, and the legacy domain became a permanent 301 on 2026-07-30.)

Prepared 2026-07-28, updated 2026-08-16 from the as-built system; companion views:
[Conceptual](#) · [Logical](#)